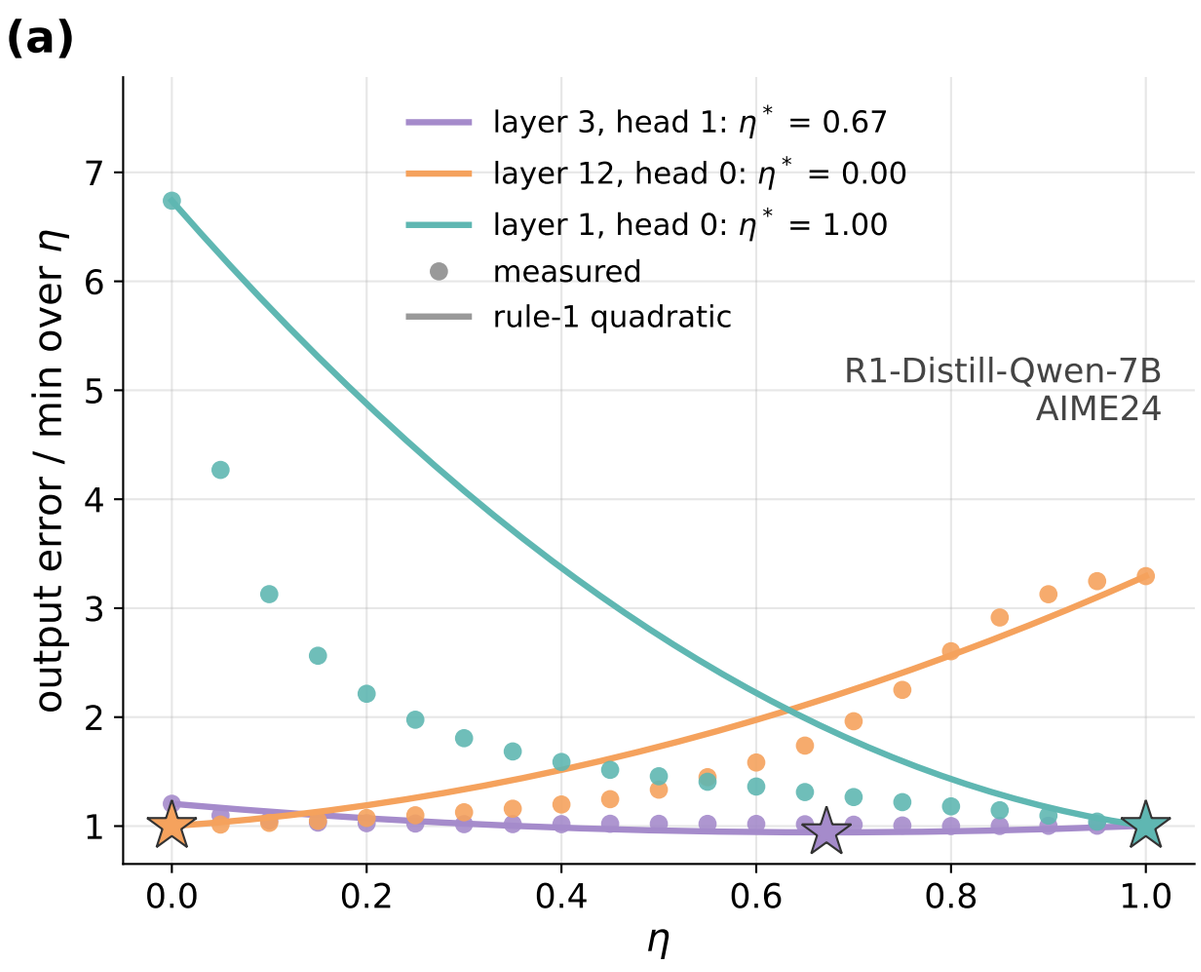
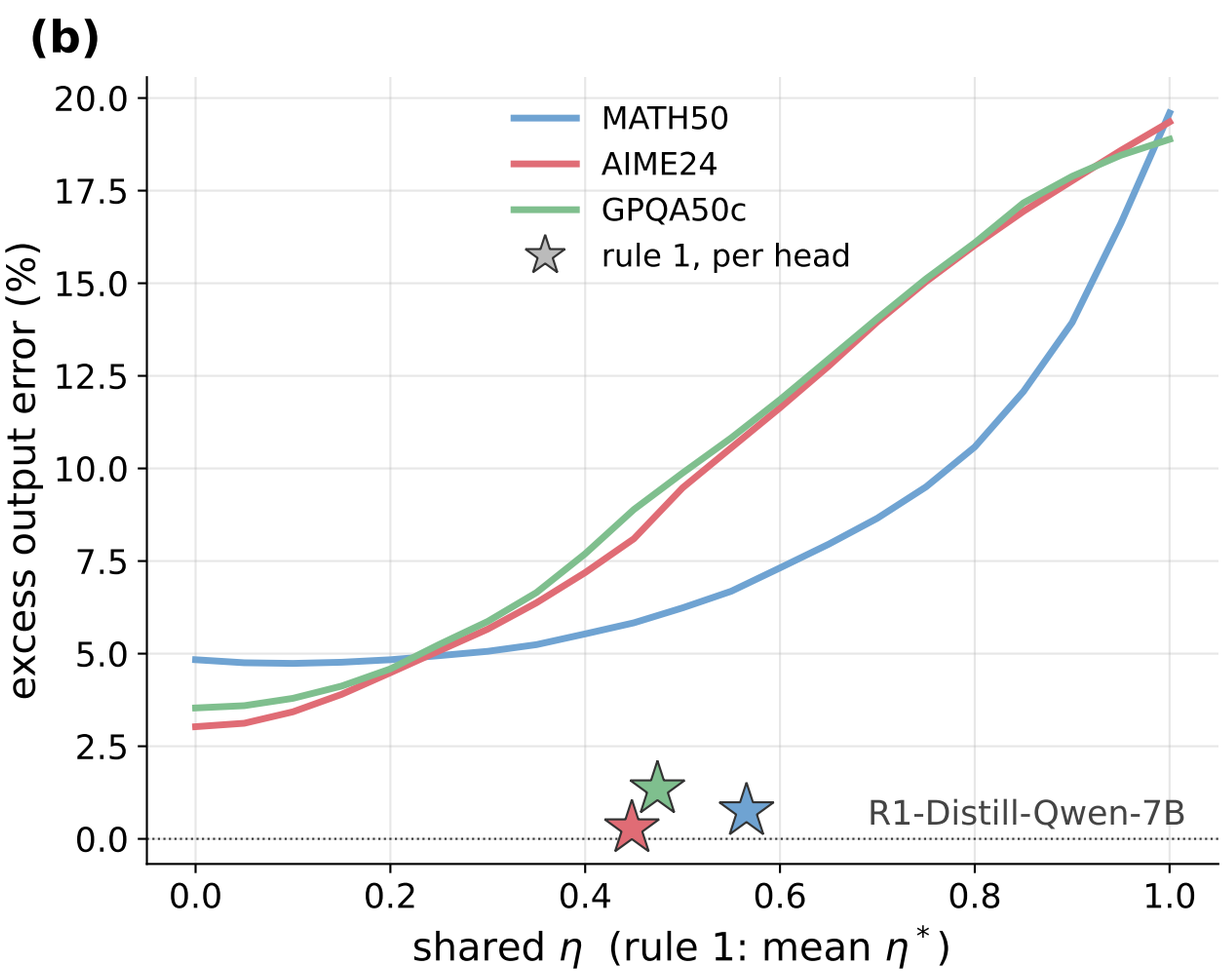
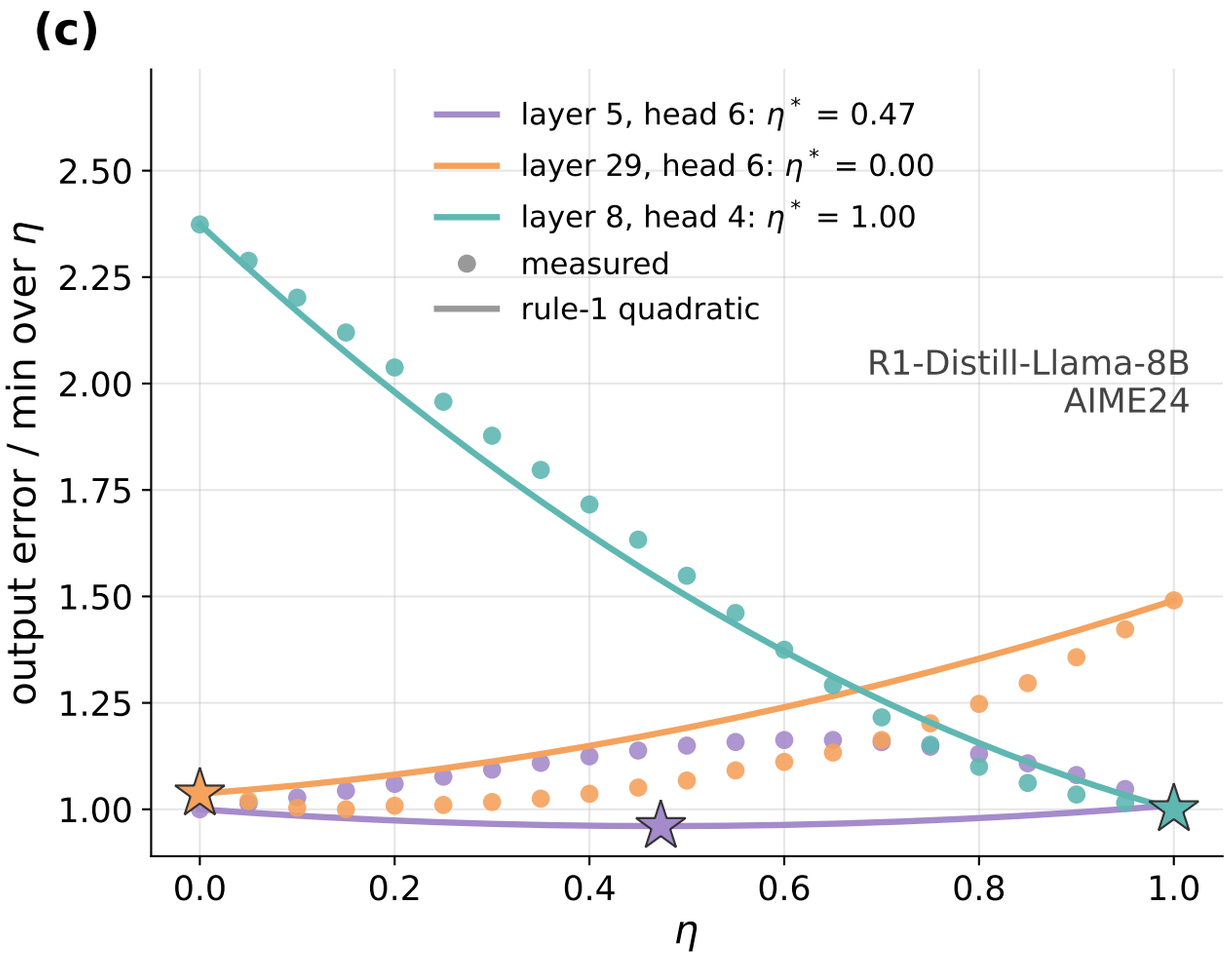
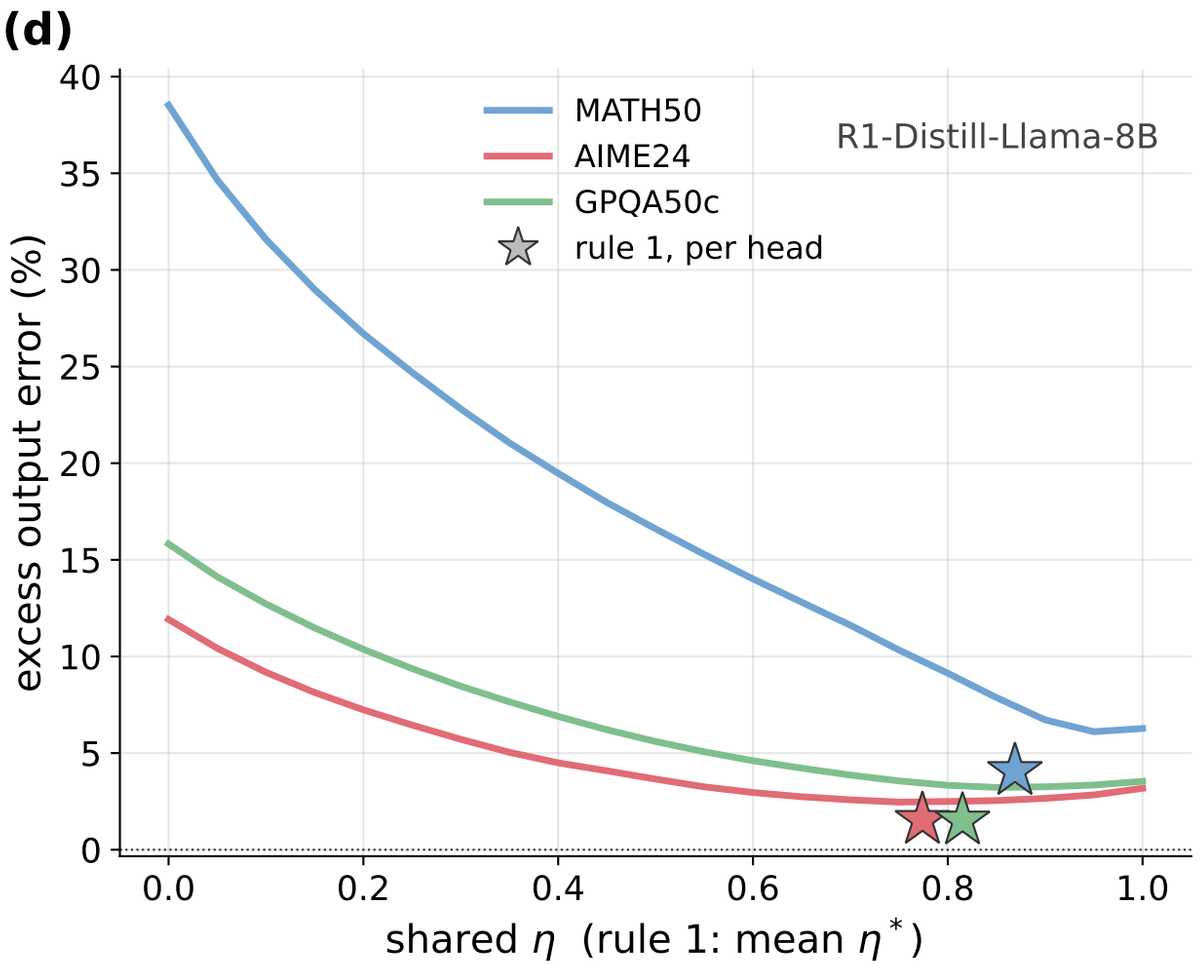

(a)
layer 3, head 1: η^* = 0.67
layer 12, head 0: η^* = 0.00
layer 1, head 0: η^* = 1.00
measured
rule-1 quadratic
R1-Distill-Qwen-7B
AIME24
output error / min over η
η


(b)
MATH50
AIME24
GPQA50c
rule 1, per head
R1-Distill-Qwen-7B
excess output error (%)
shared η (rule 1: mean η^*)


(c)
layer 5, head 6: η^* = 0.47
layer 29, head 6: η^* = 0.00
layer 8, head 4: η^* = 1.00
measured
rule-1 quadratic
R1-Distill-Llama-8B
AIME24
output error / min over η
η


(d)
MATH50
AIME24
GPQA50c
rule 1, per head
R1-Distill-Llama-8B
excess output error (%)
shared η (rule 1: mean η^*)